

Cosmic Watch: Modelling Muon Attenuation in water

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To understand the distribution of particles and their energies present in cosmic shower radiation from space, it is important to be able to characterize and separately measure different particles at different energy levels. Here, we look at how ionizing radiation is reduced at different submersion depths in water, and use the configuration of the cosmic watch device to separate out the potential muon radiation from background. We find that with an exponential decay over depth model that muons are attenuated at a rate of $\mu = 0.0345 \pm 0.002(ft^{-1})$ when submerged in the Charles River in Cambridge, MA, USA. The location was chosen as a relatively deep source of water in proximity to the researcher.

I. BACKGROUND

As the earth is bombarded by stray high energy particles in cosmic rays from outer space, the bulk of these being stray nuclei produced by supernova explosions, a number of them make it through the magnetic field shielding the planet and interact with particles in the atmosphere. These particles that experience nuclear interactions give rise to a rain of particle radiation that scatters down, and high energy components of this radiation will often make it to the surface. ⁴

Of these, we are concerned with measuring muon particles. We do this through the use of stacked scintillator sensors dubbed cosmic watches, which we can effectively filter out other potential forms of particle and photon radiation through our experimental implementation. Muons that pass through the pair of stacked sensors will produce a signal in both sensors at the same time. We will call this pair of signals in time a coincidence event, and can count these over time to get a count rate for high energy muons per second.

The signal is produced by ionization of the scintillator inside each sensor, and whether or not ionization occurs depends on the kinetic energy of the radiation. When the muon particles have in the range of $10^2 - 10^5 MeV$ of kinetic energy, they are considered to be ionizing, and should produce a signal in the detector. To understand the distribution of muon energies, we consider how passing through a dense material will reduce the kinetic energy until it is no longer ionizing.

Water, being the standard for the SI unit of density at $1000kg/m^3$, is abundant, relatively dense, and well understood as a material. Understanding how large volumes of water will attenuate the kinetic energy of muons as they pass through it provides a good standard baseline, to which materials of other densities can be easily compared.

We will measure the coincidence count rate of the cosmic watch device pair as they are lowered in a watertight vessel to different depths of the Charles River in Cambridge, MA.

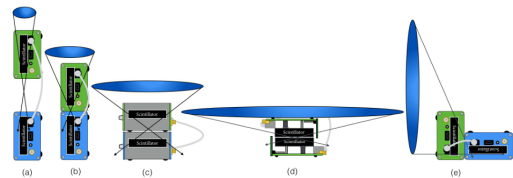


FIG. 1. Some suggested configurations of the cosmic watches in coincidence mode, allowing for the measurement restriction of incoming radiation to smaller solid angles.

II. APPARATUS

The cosmic watch is a simple self-contained apparatus that uses plastic scintillator accompanied with a silicone photomultiplier (SiPM) to measure ionizing radiation. The output of the SiPM is processed by an on-board Teensy 4.0 micro-controller, commonly and cheaply available and often used by hobby electronics enthusiasts. The cosmic watch comes with the ability to be chained to another cosmic watch, with the motivation of measuring when coincidence events occur. This allows for restriction of the solid angle through which incoming radiation is considered, and is important in limiting the measured radiation to that which can pass through both detectors in a relatively straight line. Muons will typically be the detector triggers for coincidence with small angle, since other kinds of radiation will either struggle to penetrate both watches or be deflected so wideley by the first interaction that it is unlikely they will pass through the second sensor. ¹

To submerge the cosmic watch detectors in water, we were provided with a watertight vessel by the MIT JLab staff. Two cosmic watches were removed from their protective casings, and placed inside the vessel stacked with a spacer between them to reduce the coincidence angle. A portable battery bank was then connected to the sensors and placed at the top of the stack. This vessel was then sealed ready for lowering. ^{2 3}

To lower the apparatus into the water, we used a large bucket filled partially with sand (in an attempt to keep the vessel upright) tied to a length of 40ft climbing rope.

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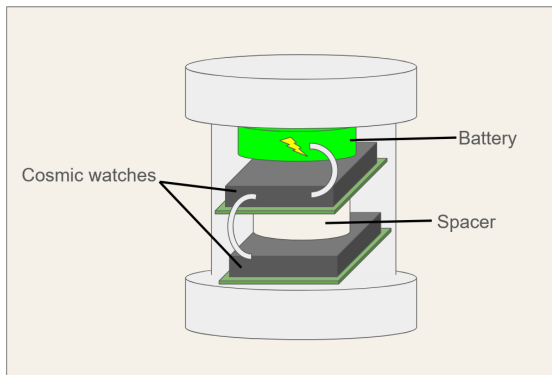


FIG. 2. Cosmic watch arrangement in watertight steel vessel.



FIG. 3. Steel submersion vessel

Once data collection was completed, the data was removed from the cosmic watches and processed in a jupyter notebook.

The model we decided to test on the data was one of exponential decay of the count rate. It is an empirical model, and a rather simple one at that. We use I_0 to denote the count rate from cosmic shower events, and I_b to represent background radiation coming from the earth below. Each of these is modeled as decaying with the amount of water shielding between each source. μ denotes our muon attenuation factor, and λ is the background attenuation.

$$I = I_0 e^{-\mu d} + I_b e^{-\lambda(D_{max}-d)} \quad (1)$$

III. METHOD

The site of data collection was the Charles River, and the measurement location was chosen by consulting the

survey map shown in figure ?? for the deepest part of the river bed. Landmark references were taken down and GPS was used when on the water to monitor location.

To get out onto the water, MIT Sailing Pavilion lent us use of a small diesel boat. Once at the site, a small anchor was dropped down, and we began data measurement by taking a reading at effective sea level on the boat for approximately 10 minutes. Then we rechecked the position to account for any drift the boat may have experienced, corrected our position and then lowered the vessel down until tension in the line was slightly relieved, indicating we had hit the bottom of the river. We recorded this depth as 33ft based on 1ft markings on the rope. we then recorded data at different depths according to the table I

In considering error for depth, note that these markings are made every foot with 1 inch tape, and in addition there is some current and tidal shift in the water that kept us bobbing between heights with a range of about 1ft.

Due to time and weather constraints, the second data set taken by Sophie Vulpe was done several days later. The second dataset was taken from the dock of the MIT Sailing pavilion boathouse, and so we expect the background count rate to be different due to the location change. The pavilion made a point that there is a concrete bed just under the pavilion as a foundation. The temperature on this day was also different, so the base muon rate is also likely different so for the purpose of analysis I have fit separate curves. Since we have potentially different background radiation on each day we can expect λ to change, but we should expect μ to be constant. The times for this data are in table II

These times are used to consider the y errorbars in the data, following standard procedure for estimating statistical error from a random poisson distribution.

Depth (ft)	Duration (s)
0	4550
33	1072
30	1026
25	960
20	900

TABLE I. Depth and Duration Data For Dataset 1

Depth (ft)	Duration (s)
0 ft	600
1 ft	305
2 ft	288
3 ft	341
5 ft	328
13 ft	596

TABLE II. Depth and Duration Data For Dataset Sophie

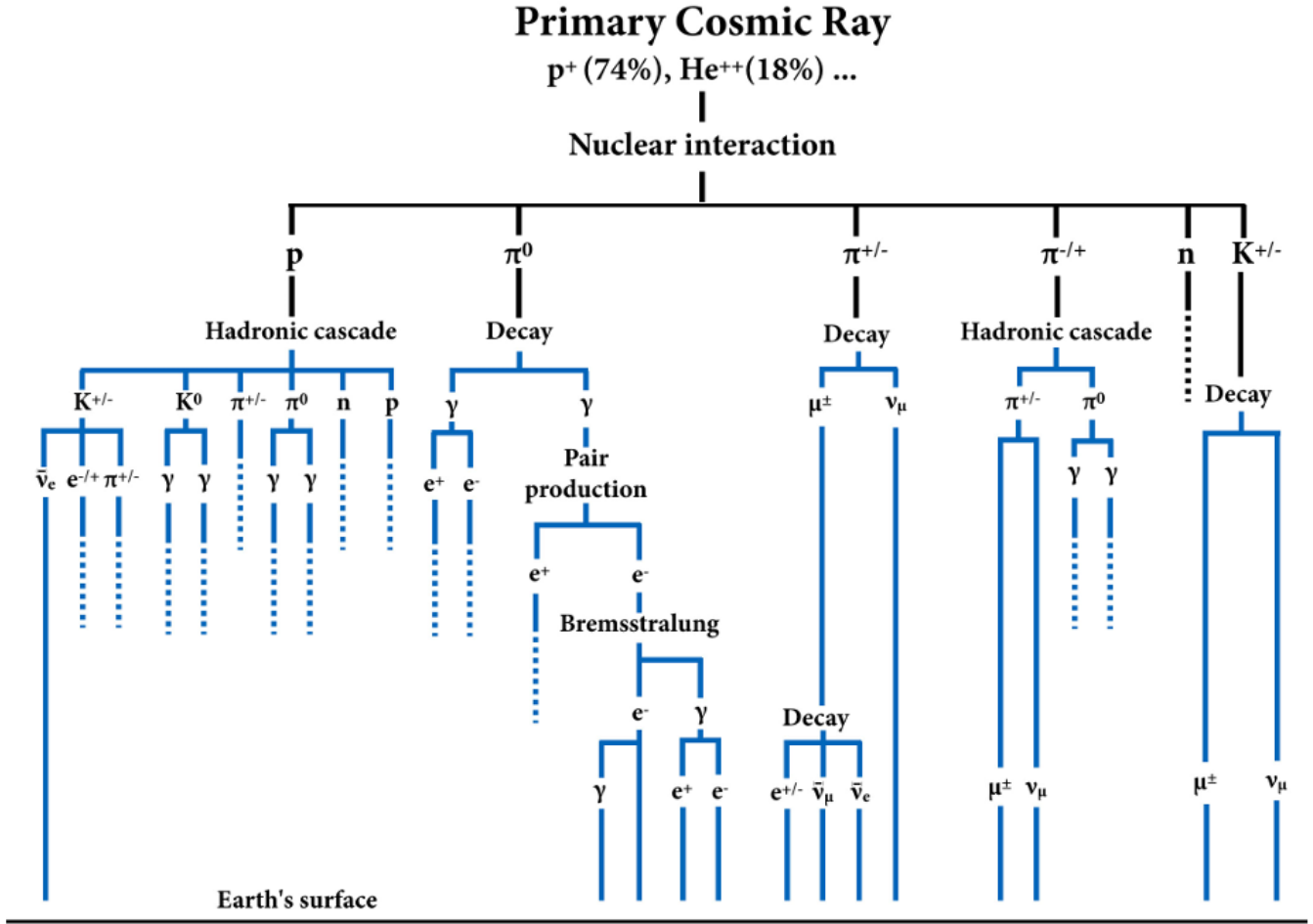


FIG. 4. Diagram of the possible decay cascades of the cosmic shower particles after nuclear interaction with the atmosphere. Diagram taken from JLab manual [2]

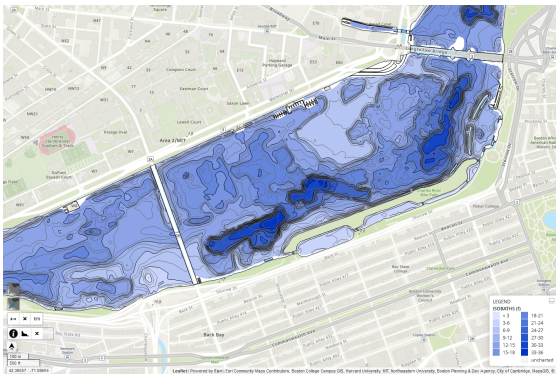


FIG. 5. Charles River survey map. Provided by the Charles River Alliance of Boaters (CRAB) in partnership with the MIT Sea Grant College Program [1]

IV. ANALYSIS

IV.1. Initial rates and coincidence counts

Binning the rate into the suitable depths based on the time of detection gives the following plots.[10][11]. Interestingly, it seems that ionizing radiation starts increasing past a certain depth. I believe this to be potentially from several effects. 1. The shielding against the background coming from below is reduced enough at these depths to provide a higher count rate. 2. The incoming muons that are particularly high energy, too high energy to ionize at sea level, have enough attenuation that they now are able to ionize the scintillator. 3. The incoming rate of muons could have increased during this interval.

With these in mind, it would be important to run the study again with a control pair of sensors kept on the boat, such that we would always have a control for sea level irrespective of time.

Regardless, the uptick should be able to be captured

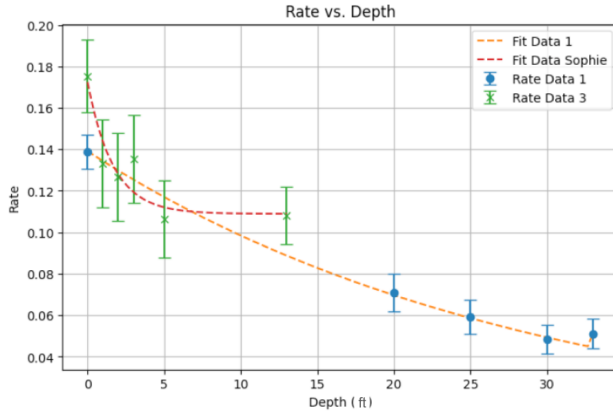
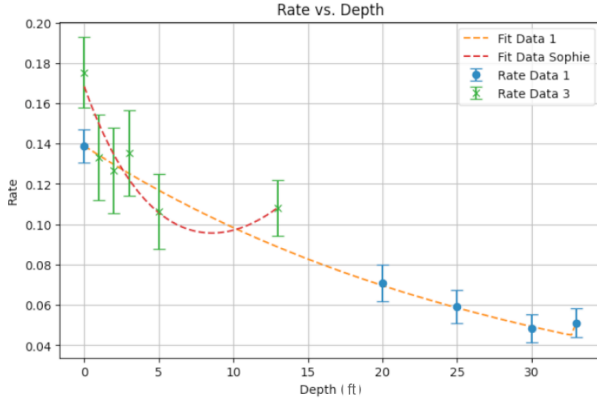


FIG. 6. Free parameter model fit

FIG. 7. Fixed parameter model fit with $I_0 = 0.160$

by our simple empirical model.

IV.2. Model fitting

The model was fitted to the data in several different ways to evaluate its robustness. The python package Scipy used to analyse this data comes with three different fitting algorithms. I also experimented with fixing the cosmic count rate to see if the attenuation coefficients would remain the same. The results of the parameters are shown below with their resulting graphs.

I_O (Hz)	I_B (Hz)	$\mu(ft^{-1})$	$\Lambda(ft^{-1})$
0.139	0.007	0.0346	9.4 ± 3
0.063	0.109	0.6 ± 0.02	0.000000001

TABLE III. Free parameter results

Observing the values given by the curve fitting, it is clear that on the separated data set cases the empirical model struggle to reliably output consistent values for either μ or λ , however when combined with the data from dataset 1 we see a μ that agrees with the rest of dataset

I_O (Hz)	I_B (Hz)	$\mu(ft^{-1})$	$\Lambda(ft^{-1})$
0.160	0.007	0.0345	9.5 ± 3
0.160	0.093	0.168	$0.08 \pm e - 7$

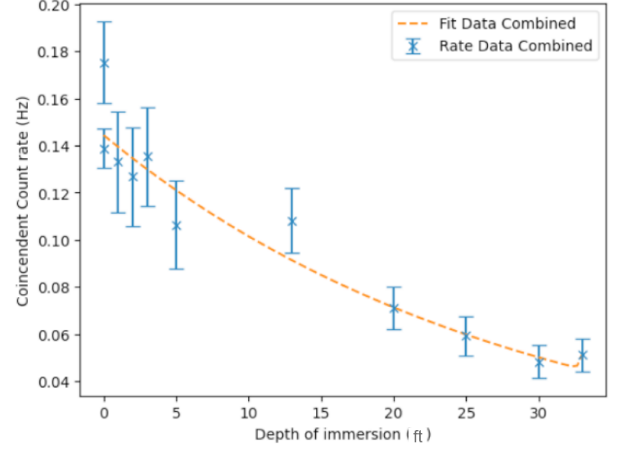
TABLE IV. Fixed I_0 parameter results

FIG. 8. Combined data fit

1s fits. As a result we make the following conclusions.

V. RESULTS

From this analysis we conclude that ionizing muons can be modelled as attenuated with $\mu = 0.0344 \pm 0.002(ft^{-1})$. Results for lambda are inconclusive, but they seem to be of order 1. Due to the difficulty in fitting, we conclude that given the sparse data and variation in possible fits, that the model is insufficiently evaluated without more controls.

There are many possible sources of error in this study. We did not control for change in muon flux overtime for example, instead assuming that it was constant. We also experienced significant drift during our deepest measurement of 33ft, where due to the drift of the boat the apparatus was dragged several feet across the riverbed, scooping up a black tar like substance into the bucket. This then remained in the bucket until the bucket was retrieved and then dumped out. It would have been interesting to take a sample to check this tar like substance for radioactive material or high count rate in the lab. Additionally as already mentioned, the datasets are from different days and different locations. The day difference could result in a change in atmospheric with different temperature and air pressure, changing the baseline muon flux. The location could also change background due to the presence of constructed buildings at the pavilion vs the riverbed.

The uptick in the graph, whilst represented in the overall count rate as well, is not too large compared to the sigma error. To be more confident of the magnitude

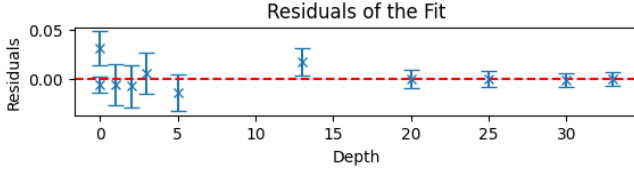


FIG. 9. Residuals for combined data fit

I_0	0.144
I_b	0.00613
λ	5.70 (covariance 1×10^{14})
μ	0.0353

TABLE V. Parameters and Values for combined data fit

of this change and the cause, we need to maintain long record times for the rates at all the depths.

VI. FURTHER WORK

To further refine and evaluate this model, I would like to take several next steps. 1. Perform longer measurements at boat house to reduce statistical errors in the findings. 2. Take a full range of depth measurements out at the center of the river to remove boathouse concrete effects, and allow for opportunity to compare. 3. Take measurements at a deeper location such as Walden Pon. 4. Characterize background of the boathouse and river using shielding above the vessel at sea level. 5. When taking submersion samples, start high and go low, such that any sediment picked up from the bottom of the riverbed isnt pulled up possibly affecting count rate. 6. Characterize riverbed sediment properties 7. Maintain pair of identical control sensors at sea level on the boat for differential control data. 8. Stack sensors side-ways for smaller solid angle limit, potentially removing

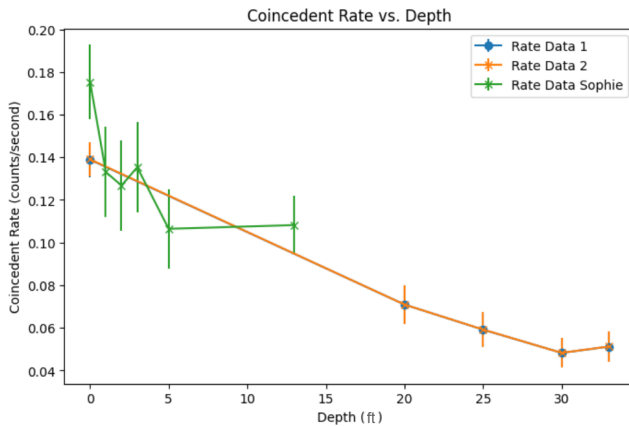


FIG. 10. Plot of coincidence counts against depth. Not the uptick at the end of both datasets.

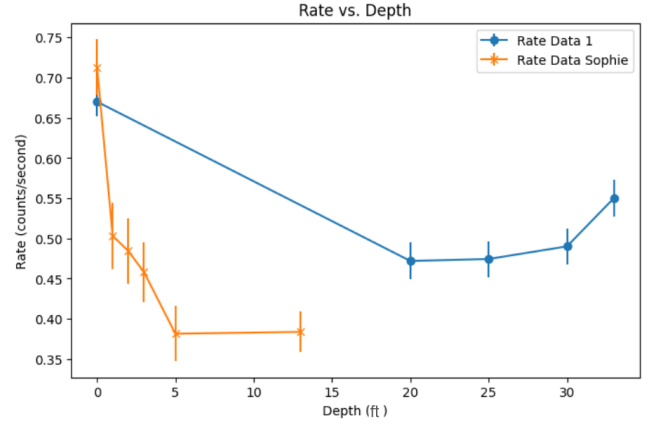


FIG. 11. Plot of raw ionization count in detector 1



FIG. 12. Lowering apparatus, a bucket and a rope.

more background radiation and compare.

VI.1. Bibliography

REFERENCES

- [1] CRAB and MIT Sea Grant Chart of the Lower Charles River. URL: <https://gisviz.mit.edu/maps/charleschart/>.
- [2] MIT Department of Physics. *8.13 Junior Lab Manual for CosmicWatch: The Desktop Muon Detector (v3)*. 2024.

VII. APPENDIX IMAGES



FIG. 13. Diesel boat with researcher Fiona Daly aboard.



FIG. 15. Sediment photo 2



FIG. 14. Sediment photo 1